33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F21-SR-001
415V CIL4 SWITCHROOM

F21-TR-007
TRANSFORMER ROOM

F21-TR-007
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

1600A
4 POLE
F21-GE-007
415V
1000kVA

F21-SW-007
CIL4 415V MOTOR CONTROL CENTRE No.2

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
Cl 1
4000/ 5A
30VA
Cl0.5
1600/ 5A
5VA
Cl 1
1600/ 1A
2.5VA
5P20
TEST
PM
AFD
ACB
00F-08
01F-08
4000A
1600A
20A
63A
2A
32A
2A
63A
240VAC SPRING
CHARGE SUPPLY
50
51
50G
51G
60G
24VDC CONTROL POWER SUPPLY
NOTE 7
NOTE 4
SD
NOTE 6
125A
160A
ATS
INC

CONTINUED ON SHT 2
2000-F21-SLD-EL-10252

EMERGENCY 415V 3NE~ 50Hz 4000A 65kA for 1sec

06F-05
05F-05
03F-05
05R-05
05R-05
06R-11
06R-18
02F-04
02F-11
630A
250A
90A
EOL
xxxA
xxxA
xx / xxA
xxVA
xPxx

E-SHUT
F21SR001-LCS
NOTE 12

RTD
M
3~
250kW
F21-AG-415
CIL4
TANK 5 AGITATOR
2000-F21-SCM-EL-10259

F21-AG-416
CIL4
TANK 6 AGITATOR
2000-F21-SCM-EL-10260

F21-AG-417
CIL4
TANK 7 AGITATOR
2000-F21-SCM-EL-10261

F21-AG-418
CIL4
TANK 8 AGITATOR
2000-F21-SCM-EL-10262

F21-AG-419
CIL4
TANK 9 AGITATOR

75kW
F21-SC-415A
CIL4 TANK 5
INTERTANK SCREEN 1
2000-F21-SCM-EL-10270

F21-SC-415B
CIL4 TANK 5
INTERTANK SCREEN 2
2000-F21-SCM-EL-10271

F21-SC-416A
CIL4 TANK 6
INTERTANK SCREEN 1
2000-F21-SCM-EL-10272

F21-SC-416B
CIL4 TANK 6
INTERTANK SCREEN 2
2000-F21-SCM-EL-10273

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER TO THAT EFFECT. CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F21-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 12)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 / E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F21-SLD-EL-10251_Rev1

Flotation tailings CIL4 F21-SW-007 415V MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F21 - FLOTATION TAILINGS CIL 4\2000-F21-SLD-EL-10251DWG 06NOV25 11:07AM HOTSEAT04